



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Bearings, Constructions & Scale Drawings

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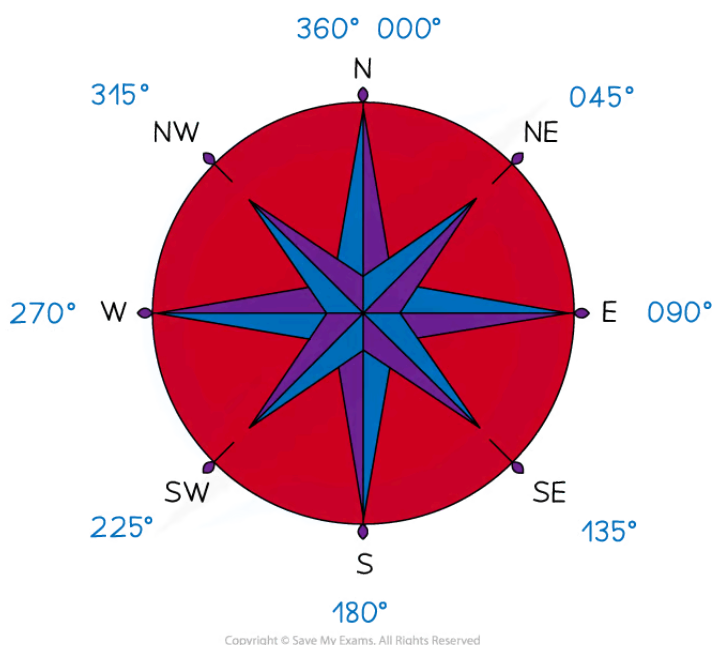
Your notes

Bearings

Bearings

What are bearings?

- **Bearings** are a way of describing an **angle**
 - They are commonly used in **navigation**
- There are **three rules** which must be followed when using a bearing:
 - They are measured **from North**
 - North is usually straight up on a scale drawing or map, and should be labelled on the diagram
 - They are measured **clockwise**
 - The angle should always be written with **3 digits**
 - 059° instead of just 59°
- Knowing the **compass directions** and their respective bearings can be helpful





Your notes

How do I find a bearing between two points?

- Identify where you need to **start**
 - "The bearing **of A from B**" means **start at B** and find the bearing to A
 - "The bearing **of B from A**" means **start at A** and find the bearing to B
- Draw a **North line** at the **starting point**
- Draw a **line between the two points**
- **Measure** the angle between the **North line** and the **line joining the points**
 - Measure **clockwise** from North
- Write the angle using **3 figures**

How do I draw a point on a bearing?

- You might be asked to plot a point that is a **given distance** from another point and on a **given bearing**
- **STEP 1**
Draw a **North line** at the point you wish to measure the bearing **from**
 - If you are given the bearing **from A to B** draw the North line at **A**
- **STEP 2**
Measure the **angle** of the bearing given **from the North line** in the **clockwise direction**
- **STEP 3**
Draw a line and add the **point B** at the **given distance**

How do I find the bearing of B from A if I know the bearing of A from B?

- If the **bearing of A from B** is **less than 180°**
 - **Add 180°** to it to find the **bearing of B from A**
- If the **bearing of A from B** is **more than 180°**
 - **Subtract 180°** from it to find the **bearing of B from A**

How do I answer trickier questions involving bearings?

- Bearings questions may involve the use of **Pythagoras** or **trigonometry** to find missing distances (lengths) and directions (angles)
 - You should always **draw a diagram** if there isn't one given



Your notes



Examiner Tips and Tricks

- Make sure you have all the **equipment** you need for your maths exams
 - A rubber and pencil sharpener can be essential as these questions are all about accuracy
 - Make sure you can see and read the markings on your ruler and protractor
- **Always** draw a big, clear diagram and annotate it, be especially careful to label the angles in the correct places!



Worked Example

A ship sets sail from the point P, as shown on the map below.

It sails on a bearing of 105° until it reaches the point Q, 70 km away. The ship then changes path and sails on a bearing of 065° for a further 35 km, where its journey finishes.

Show on the map below the point Q and the final position of the ship.



Your notes

SCALE: 1cm = 10 km



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P

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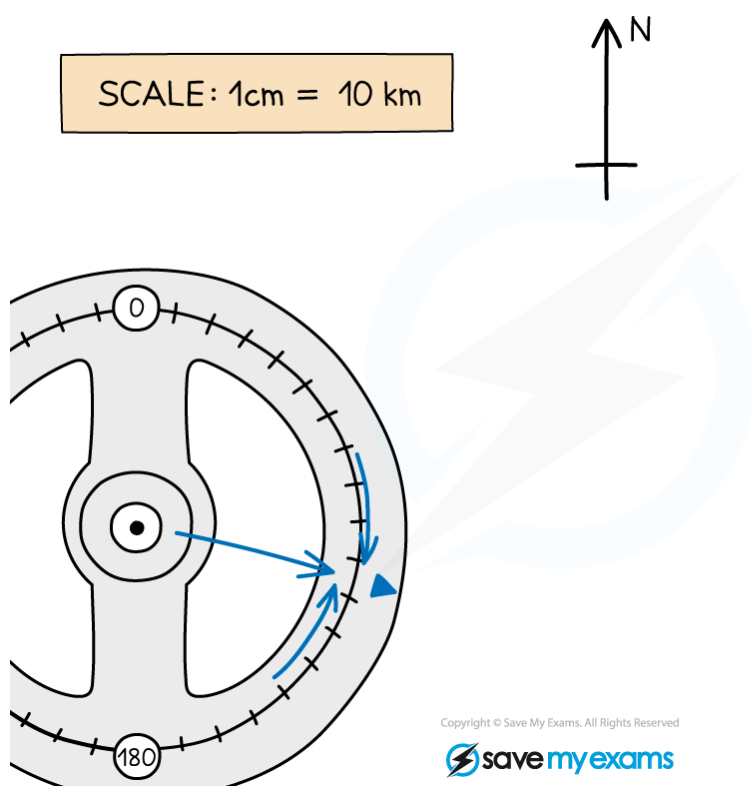
Draw in a north line at the point P

Measure an angle of 105° clockwise from the north line

Making sure you are accurate, carefully make a small but visible mark on the map



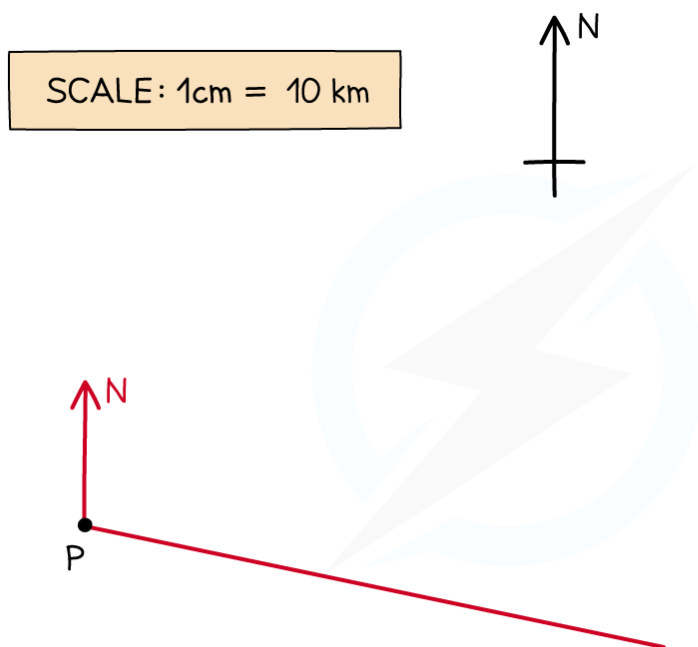
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Draw a line from P through the mark you have made. Make this line long so that you can easily measure along it accurately



Your notes



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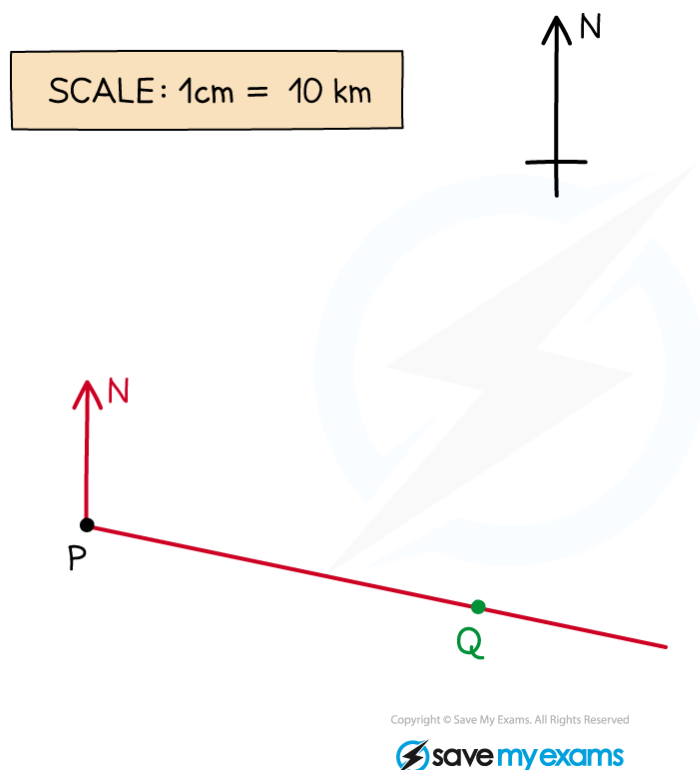
Use the scale given on the map (1 cm = 10 km) to work out the number of cm that would represent 70 km

$$70 \text{ km} = 70 \div 10 = 7 \text{ cm}$$

Accurately measure 7 cm from the point P along the line and make a clear mark on the line
Label this point Q



Your notes



A bearing of 065 means 65° clockwise from the North

First, draw a North line at the point Q, then carefully measure an angle of 65° clockwise from this line. Make a mark and then draw a line from Q through this mark

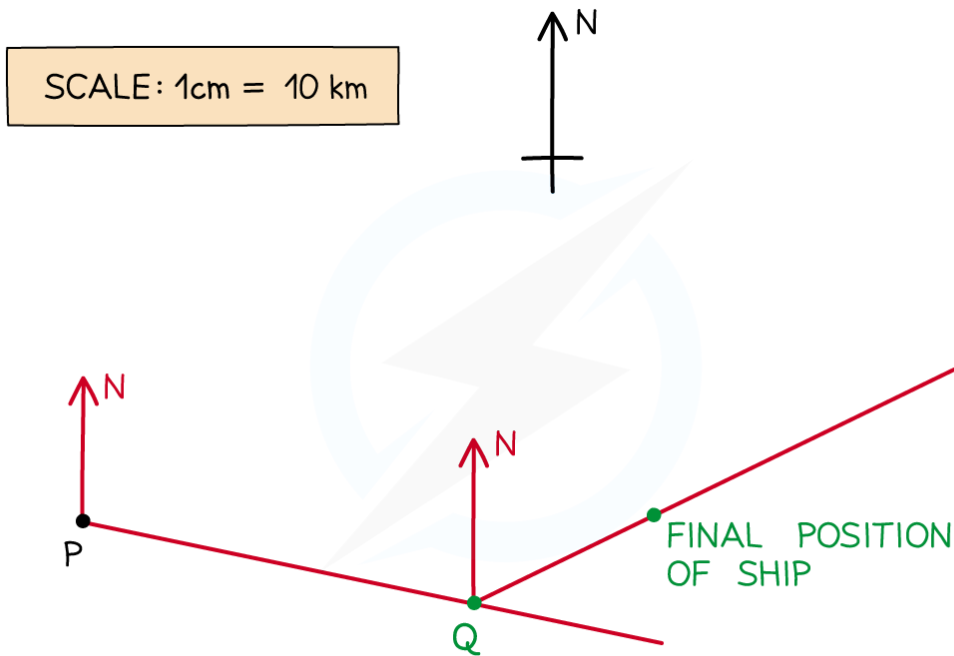
Using the scale, find the distance in cm along the line you will need to measure.

$$35 \text{ km} = 35 \div 10 = 3.5 \text{ cm}$$

Accurately measure 3.5 cm from the point Q along this new line and make a clear mark on the line. This is the final position of the ship.



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Your notes

Scale

Scale

What is a scale?

- For **accurate drawings** and **constructions** scale refers to a **ratio**
 - This ratio describes the relationship between the **drawn size** and the **real-life size**
- **Maps** are usually drawn to a scale
- The ratio will work for **any unit of length** applied to both sides
 - For example, the scale 1: 50 000 could mean 1 cm = 50 000 cm, 1 km = 50 000 km or even 1 yard = 50 000 yards
 - If you're measuring the length from a map it will be easiest to measure in cm

How can I use a scale to find the actual lengths from an accurate drawing?

- **STEP 1**
Use a ruler to **measure** the **distance accurately** on the **map**
 - For example, measuring a length from A to B as 5.8 cm
- **STEP 2**
Use the **scale** to find the **actual distance** in the **same units**
 - For example, if the scale is 1 : 150 000 the actual distance = $5.8 \text{ cm} \times 150\,000 = 870\,000 \text{ cm}$
- **STEP 3**
Convert the actual distance to a **more suitable unit**
 - For example, $870\,000 \text{ cm} = 8\,700 \text{ m} = 8.7 \text{ km}$

How can I use a scale to find lengths for an accurate drawing?

- **STEP 1**
Convert the scale into a **ratio** where one side is **1 cm** and the other side uses the **units the real distance** is measured in
 - For example, if the real distance is in km and the scale is 1 : 500 000,
 - $1 : 500\,000 = 1 \text{ cm} : 500\,000 \text{ cm} = 1 \text{ cm} : 5\,000 \text{ m} = 1 \text{ cm} : 5 \text{ km}$
 - So 1 cm on the map, represents 5 km in real life



Your notes

STEP 2Use this ratio to **convert** the **actual distance** to the **scale distance**

- For example, if the actual distance = 20 km, the scale distance will be $20 \div 5 = 4$ cm

**Examiner Tips and Tricks**

- When working with lots of different units and converting between them, make sure to use "common sense" checks
 - e.g. When converting 500 km into metres, am I expecting a bigger or smaller number?

**Worked Example**

A map is drawn where a length of 5 cm is equal to an actual distance of 0.6 km.

(a) Write the scale that is used for the map.

Convert both parts of the scale to the same units

The answer needs to be in the form $1 : n$ so convert 0.6 km into cm using $1 \text{ m} = 100 \text{ cm}$ and $1 \text{ km} = 1000 \text{ m}$

$$\begin{aligned} 0.6 \text{ km} &= 0.6 \times 1000 \text{ m} = 600 \text{ m} \\ 600 \text{ m} &= 600 \times 100 \text{ cm} = 60\,000 \text{ cm} \end{aligned}$$

Now the ratio has the same units $5 \text{ cm} : 60\,000 \text{ cm}$, you can remove the units

$$5 : 60\,000$$

Write in the form $1 : n$ by dividing both sides by 5

$$1 : 12\,000$$

(b) The width of a park on the map is 17 mm.

Find the actual width of the park, giving your answer in metres.

Convert 17 mm into cm

$$17 \text{ mm} = 1.7 \text{ cm}$$

Use the scale to find 1.7 cm on the map in real life

$$1.7 \text{ cm} \times 12\,000 = 20\,400 \text{ cm}$$



Your notes

Convert to metres

$$20\,400\text{ cm} \div 100$$

204 m

(c) The distance from the mouth of the ocean to the first bridge over a river is 125 metres.

Find this distance on the map.

Convert 125 metres to cm

$$125\text{ m} = (125 \times 100\text{ cm}) = 12\,500\text{ cm}$$

Use the scale to find 12500 cm in real life on the map

$$12\,500\text{ cm} \div 12\,000 = 1.0416\dots\text{ cm}$$

1.04 cm



Your notes

Constructing Triangles

Constructing SSS Triangles

What are SSS Triangle constructions?

- SSS just stands for **side, side, side!**
- An **SSS triangle construction** is a triangle **drawn accurately** if you are **told the lengths of its three sides** and no other information
- You will need:
 - a **pencil**
 - a **ruler**
 - a **pair of compasses**

How do I construct an SSS triangle?

- **STEP 1**
Use a **ruler** to draw the **longest** side as the **horizontal** base near the bottom of the space you have been given
 - This needs to be **accurate**
 - Write its length (with units) just underneath
- **STEP 2**
Open your **compasses** so that the **length** from the **compass point to the tip of your pencil** is **exactly the length** of one of the **remaining sides**
 - Being extra careful not to change the length, put the compass point on one end of the horizontal line you have drawn and **draw an arc** above the horizontal line
 - You can draw a full circle rather than an arc if you prefer
- **STEP 3**
Open your **compasses** so that the length from the **compass point to the tip of your pencil** is **exactly the length** of the **third side**
 - Being extra careful not to change the length, put the compass point on the other end of the horizontal line and **draw another arc**, making sure that it **crosses the first arc**
- **STEP 4**



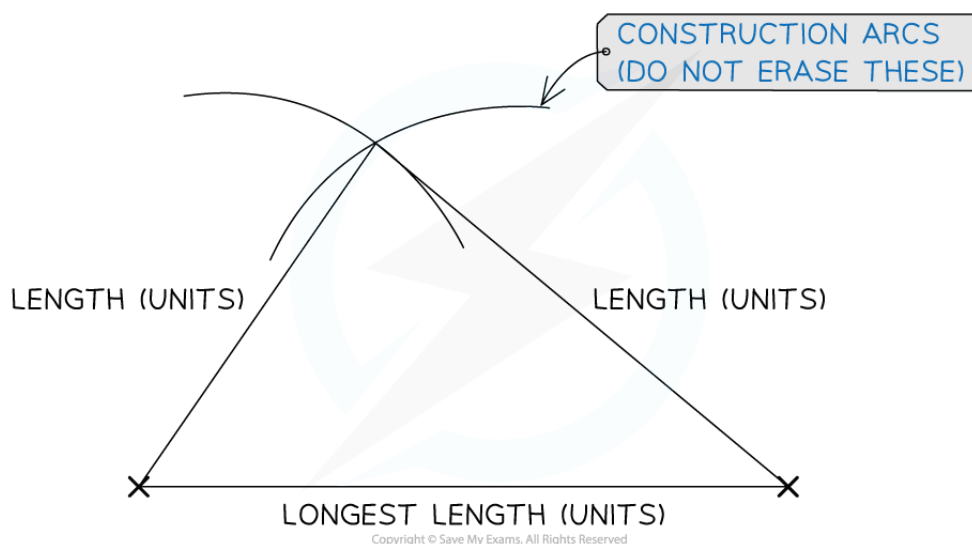
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Use your **ruler** to draw **straight lines** from each end of the horizontal line to the point **where the arcs cross**

▪ **STEP 5**

Use your **ruler** to **check** that the two new lines are **exactly equal** to the lengths given in the question

- When you are confident that they are accurate, label the lines with their lengths and units
- **Do not rub out your arcs** as the examiner will use these to check your work
- Sometimes the instructions will include a **triangle name** such as triangle ABC
 - Make sure you label each vertex with the correct letters



Examiner Tips and Tricks

- SSS triangle construction questions are usually worth 2 marks; one for having the sides accurate and the other for clearly showing the construction arcs on your diagram
- To ensure you get that second mark
 - Make sure you are confident using a pair of compasses accurately
 - Do not erase the construction arcs from your diagram



Worked Example



Your notes

Using a ruler and pair of compasses only, construct a triangle with sides 6 cm, 7 cm and 10 cm.
Do not erase your construction arcs.

Draw the 10 cm line as the horizontal base

In turn, place the point of the compasses at each end and draw an arc with radius 6 cm from one end, and another arc with radius 7 cm from the other end

The third vertex of the triangle is the point at which they intersect

Use your ruler to measure each side and check for accuracy

